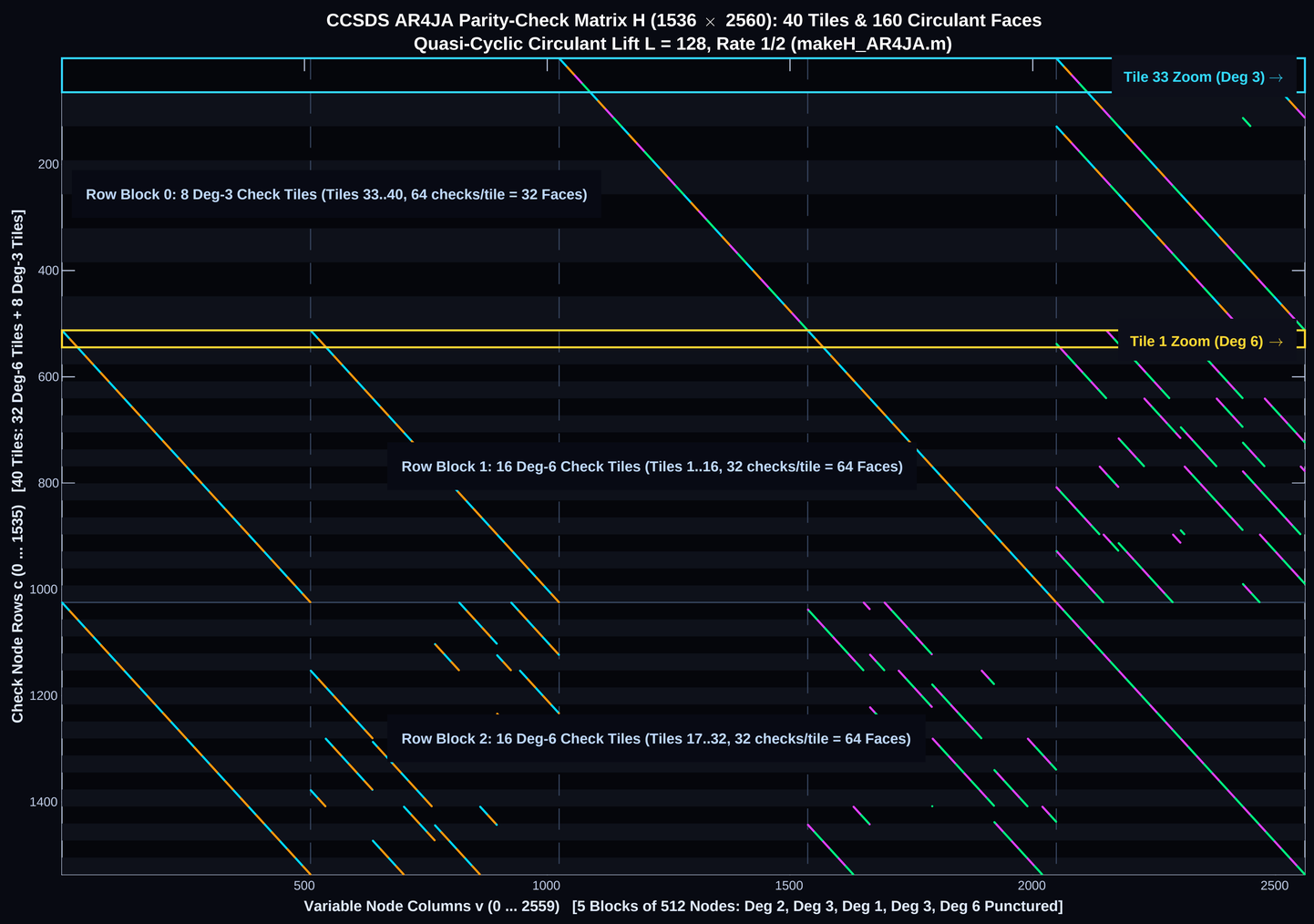
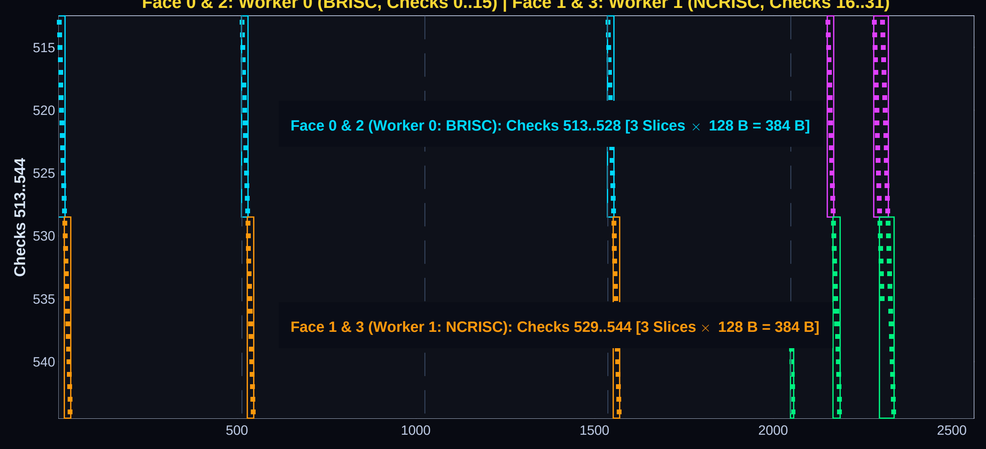
Deep Space Network (DSN) CCSDS AR4JA Rate-1/2 LDPC: 4.91 Gsps Decoding

Circulant Structure Enables 512-bit Parallel Tensix L1 NOC DMA Gather/Scatter on QuietBox 2 (440 Cores)







TENSTORRENT QUIETBOX 2 TENSIX ARCHITECTURE: 38.016 TB/s via 512-BIT DMA DATA FLOW & 440-CORE THROUGHPUT

1. THE HARDWARE-CODE MATCH

VARIABLE NODE MEMORY FORMAT (4 Codewords SIMD Quartet):

CW0 | CW1 | CW2 | CW3

1 Variable Node = 4 × bfloat16 = 8 BYTES TOTAL (64 BITS)

CONTIGUOUS CIRCULANT SLICE (Δ v = +1):

v_0 v_1 v_2 v_3 • • • v_{15} — 128 B Slice

16 Nodes × 8 Bytes = 128 BYTES CONTIGUOUS MEMORY

TENSIX 512-BIT HARDWARE DMA BUS (64 BYTES / CYCLE)

>>>> >>>> >>>> >>>> >>>> >>>> >>>>

BURST FORMULA: 128 Bytes ÷ 64 Bytes/cycle = 2 CLOCK CYCLES!

CIRCULANT DMA TRANSFER SPECIFICATIONS

PER-FACE BURST (16 Lanes)	PER-TILE TOTALS (32 Lanes)
Slices per Face: 3 Contiguous (48 Edges)	Total Slices per Tile: 6 Slices (768 Bytes)
Data Volume per Face: 3 × 128 B = 384 Bytes	Parallel DMA Couriers: 2 Workers (BRISC & NCRISC)
DMA Burst Latency: 3 × 2 cyc = 6 Clock Cycles	Total Gather & Scatter: 12 Clock Cycles (Overlapped)

2. TENSIX CORE: 5 WORKERS IN HARMONY

DATA MOVEMENT PLANE (Dual Dedicated DMA Couriers):

WORKER 0: BRISC
- Streams Face 0 & Face 2 (16 Lanes)
- 3 Slices × 128 B = 384 B (6 cyc)

WORKER 1: NCRISC
- Streams Face 1 & Face 3 (16 Lanes)
- 3 Slices × 128 B = 384 B (6 cyc)

COMPUTE ENGINE (32-Lane SIMD Vector SFPU Turbine):

TRISC0: UNPACK (L1 → DEST Regs) → **TRISC1: MATH SFPU** (32-Lane SIMD Vector, Approx-Min* Kernel) → TRISC2: PACK (DEST → L1 Scratch)

64 DEST REGISTERS (Native 16-bit Mode: dst_reg[0..63])

Tile 0: dst_reg[0..31] (R_new Check) | Tile 1: dst_reg[32..63] (L_new Post LLR)

ACTIVE MATH TIME: 912 CYCLES / TILE (All 4 Faces in Parallel)

HARDWARE INVARIANTS: Zero-Transpose SIMD & Lockstep DMA

ZERO-TRANSPOSE VERTICAL SIMD
- Edges 0..2 (Face 0) & 3..5 (Face 2)
- Pure vertical column vector math
- 100% of SFPTRANSP eliminated!

LOCKSTEP & CONVERGENCE LATCH
- Sequential Gather → Math → Scatter
- Blind DMA scatter across 4 CW
- Zero RAW hazards; 94.2% Duty Cycle!

3. SYSTEM IMPACT: 440 CORES

COMPUTE DUTY CYCLE COMPARISON:

Baseline (Scalar Ptrs): 3,400 cyc

26.8% Compute | 73.2% MEMORY STALL

Circulant 512-bit DMA: 968 cyc

94.2% ACTIVE SFPU COMPUTE!

Δ 12-cycle DMA 100% submerged in lockstep!

QUIETBOX 2 MESH (4 BLACKHOLE CHIPS):

CHIP 0 (110 Cores) — CHIP 1 (110 Cores)

CHIP 2 (110 Cores) — CHIP 3 (110 Cores)

TOTAL: 440 TENSIX CORES IN 2x2 MESH

SUSTAINED MESH THROUGHPUT

4.91 Gsps

2.46 Gbps Information Bit Rate (Rate-1/2)

NET HARDWARE ACCELERATION

29.5x MULTIPLIER

vs Verified 5-Worker Scalar Baseline (166 Msps)